

RStudio Lab Week 1 - Simple Linear Regression

Notes

A brokerage house would like to be able to predict the number of trade executions per day and has decided to use the number of incoming phone calls as an explanatory variable. Data were collected over a period of 35 days and saved in the dataset calls.xlsx. The task at hand is to perform a regression analysis on the data in order to develop a prediction equation for the number of trade executions as a function of the number of calls per day. (Note: All answers must be entered rounded to TWO decimal places).

Datasets

All questions relate to the 'calls.xlsx' data.

Instructions

- 1) Make sure that you have downloaded the dataset 'calls.xlsx'.
- 2) Copy the dataset into your student drive.
- 3) Make sure you have opened RStudio.
- 4) Set your working directory to your student drive by using Session – Set working directory – Choose directory and navigate to your student drive (or wherever you have saved the dataset for this quiz).
- 5) Insert a new R Script, using File – New File – R Script OR by clicking the white page with the green + button and selecting R Script.
- 6) Import the 'calls.xlsx' dataset. You may need to use the function **read_excel()** from the **readxl** package.
- 7) Find the mean, variance and standard deviation of each of the variables in the dataset.
- 8) Apply the **cor()** function to the data where the independent variable *X* is Calls and the dependent variable *Y* is Executions to estimate the correlation coefficient.
- 9) Use the **plot()** function to create a scatterplot with Executions on the y axis and Calls on the x axis.
- 10) Perform a simple linear regression using the **lm()** function and save results as an object called fit.
Hint: *formula = dependent variable ~ explanatory variable*
- 11) Use **summary()**, **anova()** and **confint()** on the fit object to generate the summary of regression results, the anova table and 95% confidence intervals for the coefficients.